

CRT in D-dim: $\eta_{\mu\nu} = (\underbrace{-1, \dots, -1}_S, \underbrace{1, \dots, 1}_r)$

• Conf. is a diffeomorphism $x^\mu \mapsto x'^\mu(x)$ s.t.

$$ds^2 = dx'^\mu dx'^\nu \eta_{\mu\nu} = \Omega(x) ds^2$$

Define

$$f^\mu(x) \equiv x'^\mu(x)$$

then $dx'^\mu = df^\mu = \frac{\partial f^\mu}{\partial x^\lambda} dx^\lambda$

$$\Rightarrow \frac{\partial f^\mu}{\partial x^\lambda} \frac{\partial f^\nu}{\partial x^\sigma} \eta_{\mu\nu} = \eta_{\lambda\sigma} \Omega(x) \quad \textcircled{\star}$$

$$\partial_\lambda f^\mu \partial_\sigma f^\nu \eta_{\mu\nu} \eta^{\lambda\alpha} = \delta_\sigma^\alpha \Omega(x)$$

take $\alpha = \sigma$, $\delta_\sigma^\sigma = D$

$$\frac{2}{D} \partial^\lambda f^\mu \partial_\lambda f_\mu = \Omega(x)$$

Now, consider $f^\mu(x) = x^\mu + \varepsilon^\mu(x)$

$\rightarrow$ Go to $\mathcal{O}(\varepsilon)$, drop $\mathcal{O}(\varepsilon^2)$:

$$\left(\frac{2}{D}\right) \partial^\lambda (x^\mu + \varepsilon^\mu) \partial_\lambda (x_\mu + \varepsilon_\mu) = \Omega(x)$$

$$\partial^\lambda x^\mu \partial_\lambda x_\mu + 2 \partial^\lambda \varepsilon^\mu \partial_\lambda x_\mu = \Omega(x)$$

$$\eta^{\lambda\sigma} \delta_\sigma^\mu \eta_{\mu\nu} \delta_\lambda^\nu + 2 \partial^\lambda \varepsilon^\mu \eta_{\mu\nu} \delta_\lambda^\nu = \Omega(x) D$$

$$\eta^{\lambda\sigma} \eta_{\sigma\lambda} + 2 \partial^\lambda \varepsilon^\mu \eta_{\mu\lambda} = D \Omega(x)$$

$$D + 2 \partial^\lambda \varepsilon_\lambda = D \Omega(x)$$

$\therefore$

$$\Omega(x) = \left(1 + \frac{2}{D} \partial^\lambda \varepsilon_\lambda\right)$$

Taking this and plugging it into $\textcircled{\star}$:

$$\partial_\lambda f^\mu \partial_\sigma f^\nu \eta_{\mu\nu} = \eta_{\lambda\sigma} \left(1 + \frac{2}{D} \partial^\lambda \varepsilon_\lambda\right)$$

$$\partial_\lambda x^\mu \partial_\sigma x^\nu \eta_{\mu\nu} + \partial_\lambda x^\mu \partial_\sigma \varepsilon^\nu \eta_{\mu\nu} + \partial_\lambda \varepsilon^\mu \partial_\sigma x^\nu \eta_{\mu\nu} = \eta_{\lambda\sigma} \left(1 + \frac{2}{D} \partial^\lambda \varepsilon_\lambda\right)$$

$$\delta_\lambda^\mu \delta_\sigma^\nu \eta_{\mu\nu} + \delta_\lambda^\mu \partial_\sigma \varepsilon_\mu + \partial_\lambda \varepsilon_\nu \delta_\sigma^\nu = \eta_{\lambda\sigma} \left(1 + \frac{2}{D} \partial^\lambda \varepsilon_\lambda\right)$$

$$\eta_{\lambda\sigma} + \partial_\sigma \varepsilon_\lambda + \partial_\lambda \varepsilon_\sigma = \eta_{\lambda\sigma} + \eta_{\lambda\sigma} \left(\frac{2}{D}\right) \partial^\mu \varepsilon_\mu$$

We therefore conclude

$$\partial_\sigma \varepsilon_\lambda + \partial_\lambda \varepsilon_\sigma = \eta_{\lambda\sigma} \left(\frac{2}{D}\right) \partial^\mu \varepsilon_\mu$$

• Taking the ∂^λ derivative we find:

$$\partial^\lambda \partial_\sigma \xi_\lambda + \partial^\sigma \xi_\sigma = \frac{2}{D} \partial_\sigma (\partial^\lambda \xi_\lambda)$$

$$(1 - \frac{2}{D}) \partial_\sigma (\partial^\lambda \xi_\lambda) + \partial^\lambda \partial_\lambda \xi_\sigma = 0$$

$$(D-2) \partial_\sigma (\partial^\lambda \xi_\lambda) + D \partial^\lambda \partial_\lambda \xi_\sigma = 0$$

$$* (2-D) \partial_\sigma (\partial^\lambda \xi_\lambda) = D \partial^\sigma \xi_\sigma *$$

• Take ∂^σ : for $(D \neq 2)$

$$(2-D) \partial^\sigma (\partial^\lambda \xi_\lambda) = D (\partial^\sigma \partial^\lambda \xi_\lambda)$$

For $D=2$ this also vanishes so

$$* \partial^\sigma (\partial^\lambda \xi_\lambda) = 0$$

$$\Rightarrow (-D+2) \partial_\sigma (\partial^\lambda \xi_\lambda) = D \partial^\sigma \xi_\sigma$$

$$(2-D) \partial_\rho \partial_\sigma (\partial^\lambda \xi_\lambda) = D \partial_\rho \partial^\sigma \xi_\sigma$$

$$\partial_\rho \partial_\sigma (\partial^\lambda \xi_\lambda) = \frac{D}{2-D} \partial_\rho \partial^\sigma \xi_\sigma$$

Note, the LHS is symm in $(\rho\sigma)$ and so

$$\partial_\sigma \partial_\rho (\partial^\lambda \xi_\lambda) = \frac{D}{2-D} \partial_\sigma \partial^\sigma \xi_\rho$$

$\rightarrow$ Add the two

$$2 \partial_\rho \partial_\sigma (\partial^\lambda \xi_\lambda) = \left(\frac{D}{2-D} \right) \partial^\sigma (\partial_\sigma \xi_\rho + \partial_\rho \xi_\sigma)$$

$$= \left(\frac{D}{2-D} \right) \partial^\sigma \left[\left(\frac{2}{D} \right) \eta_{\mu\nu} \partial^\mu \xi_\nu \right]$$

But we concluded $\partial^\sigma (\partial^\lambda \xi_\lambda) = 0$, so

$$\partial_\rho \partial_\sigma (\partial^\sigma \xi_\sigma) = 0$$

$$\Rightarrow \partial_\rho \partial_\lambda \partial_\kappa \xi_\mu = -\partial_\rho \partial_\kappa \partial_\lambda \xi_\mu + \frac{2}{D} \eta_{\kappa\lambda} \partial_\rho \partial^\sigma \xi_\sigma$$

$$= -\partial_\rho \partial_\kappa \partial_\lambda \xi_\mu$$

$$\therefore \partial_\rho \partial_\lambda \partial_\kappa \xi_\mu = 0$$

Thus, ξ_μ is at most of order $(x^\mu)^2$:

$$\xi^\mu = x^\mu + a^\mu + t^\mu_\nu x^\nu + \ell^\mu_{\nu\rho} x^\nu x^\rho$$

$$\partial^\lambda \xi^\mu = \delta^{\mu\lambda} + t^\mu_\nu \delta^{\lambda\nu} + 2 \ell^\mu_{\nu\rho} x^\rho \delta^{\lambda\nu}$$

$$\partial^\mu \xi^\lambda = \delta^{\mu\lambda} + t_N^\lambda \delta^{\mu N} + 2\ell_{np}^\lambda \chi^p \delta^{\mu N}$$

$\therefore$

$$2\eta^{\mu\lambda} + t_N^\mu \delta^{\lambda N} + t_N^\lambda \delta^{\mu N} + 2\ell_{np}^\mu \chi^p \delta^{\lambda N} + 2\ell_{np}^\lambda \chi^p \delta^{\mu N}$$

$$= \eta^{\mu\lambda} \left(\frac{2}{D} \right) (D + t_N^\mu + 2\ell_{np}^\mu \chi^p)$$

Matching orders we find

$$\eta^{\mu\lambda} \left(\frac{2}{D} \right) t_\sigma^\sigma = t_N^\mu \eta^{\lambda N} + t_N^\lambda \eta^{\mu N} \quad (i)$$

$$\delta_p^\lambda \left(\frac{2}{D} \right) t_\sigma^\sigma = t_{pN} \eta^{\lambda N} + t_N^\lambda \delta_p^N$$

$$\eta_{\lambda\mu} \delta_p^\lambda \left(\frac{2}{D} \right) t_\sigma^\sigma = t_{pN} \delta_\mu^N + t_{\mu N} \delta_p^N$$

$$\boxed{\eta_{p\mu} \left(\frac{2}{D} \right) t_\sigma^\sigma = t_{pN} + t_{\mu N}}$$

$$(ii) \ell_{np}^\mu \chi^p \eta^{\lambda N} + \ell_{np}^\lambda \chi^p \eta^{\mu N} = \eta^{\mu\lambda} \left(\frac{2}{D} \right) \ell_{np}^\mu \chi^p$$

$$\partial_\sigma () = \partial_\sigma$$

$$\ell_{np}^\mu \eta^{\lambda N} + \ell_{np}^\lambda \eta^{\mu N} = \eta^{\mu\lambda} \left(\frac{2}{D} \right) \ell_{np}^\mu$$

$$\ell_{np}^\mu \eta^{\lambda N} + \ell_{np}^\lambda \delta_\mu^N = \delta_\mu^\lambda \left(\frac{2}{D} \right) \ell_{np}^\mu$$

$$\ell_{np}^\mu \delta_\lambda^N + \ell_{\lambda p}^\mu \delta_n^N = \eta_{\lambda n} \left(\frac{2}{D} \right) \ell_{np}^\mu$$

$$\boxed{\ell_{n\lambda}^\mu + \ell_{\lambda p}^\mu = \eta_{n\lambda} \left(\frac{2}{D} \right) \ell_{np}^\mu}$$

So, we conclude:

$$\boxed{t_{pN} + t_{\mu N} = \eta_{p\mu} \left(\frac{2}{D} \right) t_\sigma^\sigma} \quad (i')$$

$$\boxed{\ell_{n\lambda}^\mu + \ell_{\lambda p}^\mu = \eta_{\lambda n} \left(\frac{2}{D} \right) \ell_{np}^\mu} \quad (ii')$$

(i): Solved via

$$\boxed{t_{\mu N} = \omega_{\mu N} + \lambda \eta_{\mu N}}$$

$$\omega_{\mu N} = -\omega_{N\mu}$$

(ii) To solve we consider cyclic permutations:

$$(\ell_{np}^\mu + \ell_{\mu p}^\mu) + (\ell_{p\mu}^\mu + \ell_{\mu p}^\mu) - (\ell_{p\mu}^\mu + \ell_{np}^\mu)$$

$$\left(\frac{2}{D} \right) (\eta_{\mu N} \ell_{\sigma p}^\sigma + \eta_{p\mu} \ell_{\sigma N}^\sigma - \eta_{np} \ell_{\sigma p}^\sigma)$$

* By construction: $\ell_{np}^\mu \chi^p \chi^p$ is symmetric

$$L_{\mu\nu\rho} = 2 l_{\mu\nu\rho}$$

$$2 l_{\mu\nu\rho} = \frac{2}{D} (\eta_{\mu\sigma} l_{\nu\rho}^{\sigma} + \eta_{\rho\sigma} l_{\mu\nu}^{\sigma} - \eta_{\nu\rho} l_{\mu\sigma}^{\sigma})$$

Raise μ index:

$$l_{\nu\rho}^{\mu} = \frac{2}{D} (\delta_{\nu}^{\mu} l_{\sigma\rho}^{\sigma} + \delta_{\rho}^{\mu} l_{\sigma\nu}^{\sigma} - \eta_{\nu\rho} \eta^{\lambda\mu} l_{\sigma\lambda}^{\sigma})$$

-> Raise μ on $t_{\mu\nu}$

$$t_{\nu}^{\mu} = \omega_{\nu}^{\mu} + \lambda \delta_{\nu}^{\mu}$$

* Plugging these Back in to $f^{\mu}(x)$:

$$\begin{aligned} f^{\mu}(x) &= x^{\mu} + a^{\mu} + (\omega_{\nu}^{\mu} + \lambda \delta_{\nu}^{\mu}) x^{\nu} + \\ &\quad \frac{1}{D} (\delta_{\nu}^{\mu} l_{\sigma\rho}^{\sigma} + \delta_{\rho}^{\mu} l_{\sigma\nu}^{\sigma} - \eta^{\mu\nu} \eta_{\lambda\rho} l_{\sigma\lambda}^{\sigma}) x^{\nu} x^{\rho} \\ &= x^{\mu} + a^{\mu} + \omega_{\nu}^{\mu} x^{\nu} + \lambda x^{\mu} + \frac{2}{D} x^{\mu} l_{\sigma\rho}^{\sigma} x^{\rho} + \frac{2}{D} x^{\mu} l_{\sigma\nu}^{\sigma} x^{\nu} \\ &\quad - \frac{1}{D} x^2 \eta^{\lambda\mu} l_{\sigma\lambda}^{\sigma} \end{aligned}$$

* Define:



$$\lambda^{\mu} \equiv -\frac{1}{D} l_{\sigma}^{\sigma\mu}$$



$$f^{\mu}(x) = x^{\mu} + a^{\mu} + \omega_{\nu}^{\mu} x^{\nu} + \lambda x^{\mu} - 2 x^{\mu} \lambda^{\rho} x_{\rho} + x^2 \lambda^{\mu}$$

or

$$f^{\mu}(x) = x^{\mu} + a^{\mu} + \omega_{\nu}^{\mu} x^{\nu} + \lambda x^{\mu} + \lambda^{\mu} x^2 - 2 x^{\mu} \lambda^{\rho} x_{\rho}$$

a^{μ} : translation

ω_{ν}^{μ} : Lorentz

λ : Dilation

λ^{μ} : Special Conformal trans. (SCT)

• Counting the # of Generators:

$$D + \frac{D}{2}(D-1) + 1 + D = \frac{(D-1)(D+2)}{2}$$

Q: Why D for λ^{μ} ? last two terms can be

rewritten as

$$\lambda^\sigma (\eta^{\mu\sigma} x^\mu x_\nu - 2 x^\sigma x^\mu)$$

Leads to the Conformal Algebra:

$$\text{Conf}(\mathbb{R}^{1,5}) \cong \text{SO}(r+1, s+1)$$

↖ locally

$$\begin{aligned} [D, K_\mu] &= K_\mu, \quad [D, P_\mu] = -P_\mu, \quad [K_\mu, K_\nu] = 0 \\ [K_\mu, J_{\rho\sigma}] &= \eta_{\mu\rho} K_\sigma - \eta_{\mu\sigma} K_\rho \\ [K_\mu, P_\nu] &= -2\eta_{\mu\nu} D + 2\delta_{\mu\nu} \end{aligned}$$

Note that:

(i) Dilation: $\exp(\lambda x^\mu \partial_\mu)$ so, $D = x^\mu \partial_\mu$

(ii) For SCT's: $\frac{x'^\mu}{x'^2} = \frac{x^\mu}{x^2} + \lambda^\mu$

Can solve for x'^μ :

$$x^2 = x'^\mu x_\mu + x^2 \lambda^\mu x'_\mu \dots$$

$$x'^\mu = \frac{x^\mu + \lambda^\mu x^2}{(1 + 2\lambda \cdot x + \lambda^2 x^2)}$$

* For $\eta_{\mu\nu} \rightarrow g_{\mu\nu}$ we obtain General Coordinate transformations which scale the metric.

Conformal Transf. in Two Dim:

* We did not consider $D=2$ $\mathbb{R}^{1,1} \cong \mathbb{M}^2$

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = \eta_{\mu\nu} \partial_\rho \xi^\rho$$

$$\begin{aligned} \underline{\text{so:}} \quad 2\partial_0 \xi_0 &= -1 (\partial_0 \xi_0 + \partial_1 \xi_1) \\ &= (-1) (-\partial_0 \xi_0 + \partial_1 \xi_1) \end{aligned}$$

$$2\partial_0 \xi_0 = \partial_0 \xi_0 - \partial_1 \xi_1$$

$$\therefore \partial_0 \xi_0 + \partial_1 \xi_1 = 0$$

01 :

$$\partial_1 \xi_0 + \partial_0 \xi_1 = 0$$

If we go to Euclidean space

$$t \rightarrow (i t) = x^2$$

$$22: 2 \partial_2 \xi_2 = 1 (\partial^2 \xi_2 + \partial^1 \xi_1)$$

$$\therefore \partial_1 \xi_1 - \partial_2 \xi_2 = 0$$

$$\partial_1 \xi_2 + \partial_2 \xi_1 = 0$$

$$\begin{aligned} \text{In } \mathbb{M}^2: \partial_0^2 \xi_1 &= -\partial_1 (\partial_0 \xi_0) \\ &= +\partial_1 (\partial_1 \xi_1) \end{aligned}$$

$$\therefore (-\partial_0^2 + \partial_1^2) \xi_1 = 0$$

$$\xi_m = (\partial_0 \epsilon, -\partial_1 \epsilon) \Leftrightarrow (\partial_0^2 - \partial_1^2) \epsilon = 0$$

* ϵ is the sum of L and R hand movers

$$x^+ \equiv x^0 + x^1$$

$$x^- \equiv x^0 - x^1$$

$$\xi_m(x) = f(x^+) + g(x^-)$$

$\leadsto$

$$x^+ \rightarrow x^{+'} = f(x^+)$$

$$x^- \rightarrow x^{-'} = g(x^-)$$

$$ds^2 = -(dx^0)^2 + (dx^1)^2 = -dx^+ dx^-$$

* In Euclidean Space

$$\partial_1 \xi_2 = -\partial_2 \xi_1, \quad \partial_1 \xi_1 - \partial_2 \xi_2 = 0$$

Are the Cauchy-Riemann Eqs associated with

$$\xi = \xi_1 + i \xi_2$$

$$Z = x^1 + i x^2 \quad \bar{Z} = x^1 - i x^2$$

$$ds^2 = (dx^1)^2 + (dx^2)^2 = dZ d\bar{Z}$$

$$\partial \equiv \partial_z = \frac{1}{2} (\partial_1 - i \partial_2)$$

$$\bar{\partial} \equiv \partial_{\bar{z}} = \frac{1}{2}(\partial_1 + i\partial_2)$$

$$d^2x = dx^1 \wedge dx^2 = \frac{i}{2} dz d\bar{z} \equiv \frac{i}{2} dz \wedge d\bar{z}$$

$$\mathbb{M}^2 \rightarrow \mathbb{E}^2 \quad \hookrightarrow \begin{matrix} x^0 \mapsto -ix^1 \\ x^1 \mapsto x^2 \end{matrix} \quad \left(\begin{matrix} \text{Wick} \\ \text{Rotation} \end{matrix} \right)$$

$$x^+ = -ix^1 + x^2 \Rightarrow ix^+ = x^0 + ix^2 = z$$

$$\therefore \boxed{x^+ = -iz}$$

$$x^- = x^0 - x^1 = -ix^1 - x^2 \Rightarrow ix^- = x^1 - ix^2 = \bar{z}$$

$$\boxed{x^- = -i\bar{z}}$$

$$\begin{matrix} \partial_+ \rightarrow i\partial \\ \partial_- \rightarrow i\bar{\partial} \end{matrix}$$

* We now work in $\mathbb{E}^2 \cong \mathbb{C}$

$$ds^2 = dz \wedge d\bar{z} = \Omega(z, \bar{z}) dz' \wedge d\bar{z}'$$

$$x^+ \rightarrow x^{+'} = f(x^+)$$

$$x^- \rightarrow x^{-'} = g(x^-)$$

Become

$$z \rightarrow z' = f(z)$$

$$\bar{z} \rightarrow \bar{z}' = g(\bar{z})$$

$$\Omega = f'(z)g'(\bar{z})$$

$\Rightarrow z \rightarrow h(\bar{z})$ and $\bar{z} \rightarrow k(z)$ change the orientation of $\mathbb{E}^2$ so we do not consider them.

* let us look at Mf. transfr's

$$z' = f(z) = \sum_{n=-\infty}^{\infty} a_n z^n = z + \sum_{n \neq 0} a_n z^n$$

$$\begin{aligned} dz' \wedge d\bar{z}' &= f(z) g'(\bar{z}) dz \wedge d\bar{z} \\ df \wedge d\bar{g} &= \alpha(z) \beta(\bar{z}) dz \wedge d\bar{z} \\ \frac{\partial f}{\partial z} \frac{\partial g}{\partial \bar{z}} \end{aligned}$$

$$\therefore \frac{\partial f}{\partial z} = \alpha(z)$$

$$f(z) = z + \varepsilon(z) = z + \sum_{n \in \mathbb{Z}} a_n (z^{n+1})$$

$$\therefore \boxed{L_n = z^{n+1} \frac{\partial}{\partial z}} \quad \bigg| \quad \boxed{\bar{L}_n = \bar{z}^{n+1} \frac{\partial}{\partial \bar{z}}}$$

$$[L_n, L_m] = (n-m) L_{n+m}$$

$$[L_n, \bar{L}_m] = 0$$

$$[\bar{L}_n, L_m] = (n-m) \bar{L}_{n+m}$$

* To arrive we must do passive transf. on scalar fields

$$\varphi(z) \mapsto \varphi'(z) = \varphi(f(z))$$

$$\delta \varphi = \varphi'(z) - \varphi(z) = \varphi(z + \varepsilon(z)) - \varphi(z)$$

$$= \varphi(z) + \varepsilon(z) \partial \varphi(z) - \varphi(z) = \varepsilon(z) \partial \varphi$$

$$\therefore [\delta_1, \delta_2] \varphi = \delta_1(\varepsilon_2 \partial \varphi) - (1 \leftrightarrow 2)$$

$$= \varepsilon_2 \partial(\varepsilon_1 \partial \varphi) - (1 \leftrightarrow 2)$$

$$= \varepsilon_2 \partial \varepsilon_1 \partial \varphi + \varepsilon_2 \varepsilon_1 \partial^2 \varphi - \varepsilon_1 \varepsilon_2 \partial^2 \varphi - \varepsilon_1 \partial \varepsilon_2 \partial \varphi$$

$$= (\varepsilon_2 \partial \varepsilon_1 - \varepsilon_1 \partial \varepsilon_2) \partial \varphi$$

$$= a_n z^{n+1} \partial (a_m \bar{z}^{m+1}) - (1 \leftrightarrow 2) \partial \varphi$$

$$= a_n a_m (z^{n+m+1} (m-1) - z^{m+n+1} (n-1))$$

$$= a_n a_m z^{n+m+1} (m-n) \partial$$

$$[L_m, L_n] = L_{n+m} (m-n)$$

$$\therefore [L_n, L_m] = L_{n+m} (n-m)$$

* The 2D Conf. Alg. is infinite dim and is the direct sum of two identical algebras V

$$\text{Conf}(\mathbb{A}^2) \cong V \oplus \bar{V}$$

$$V: \text{Virasoro Alg}$$

$$\{L_0, L_{\pm 1}\} \cong \text{SL}(2, \mathbb{R})$$

* In general we are working w/ a space which contains a line element of the form

$$e^{\phi(z, \bar{z})} dz d\bar{z}$$

$$ds^2 = g_{zz} dz dz + g_{z\bar{z}} dz d\bar{z} + \dots + g_{\bar{z}\bar{z}} d\bar{z} d\bar{z}$$

$$g_{zz} = g_{\bar{z}\bar{z}} = 0, \quad g_{z\bar{z}} = g_{\bar{z}z} = \frac{1}{2} e^{\phi}$$

$$g_{\mu\nu}(z, \bar{z}) = \begin{pmatrix} 0 & \frac{1}{2} e^{\phi} \\ \frac{1}{2} e^{\phi} & 0 \end{pmatrix}$$

$$g^{\mu\nu}(z, \bar{z}) = \begin{pmatrix} 0 & 2e^{-\phi} \\ 2e^{-\phi} & 0 \end{pmatrix}$$

Take a general tensor:

$$\underbrace{A_{\underbrace{z \dots z}_m \underbrace{\bar{z} \dots \bar{z}}_{\bar{m}}}}_{\substack{\underbrace{}_n \\ \underbrace{}_{\bar{n}}}} \equiv A(z, \bar{z})$$

$$\bullet \quad z \mapsto w(z), \quad \bar{z} \mapsto \bar{w}(\bar{z})$$

$$\text{Note } A^{\mu}(x') = \frac{\partial x'^{\mu}}{\partial x^{\sigma}} A^{\sigma}(x)$$

$$\text{Thus, } A^z(w(z)) = \frac{\partial w^z}{\partial z^{\mu}} A^{\mu}(z)$$

$$\mu = z, \bar{z} \text{ but } \bar{\partial} w = 0, \text{ so}$$

$$A^z(w(z)) = \frac{dw}{dz} A^z(z)$$

likewise,

$$A_{\bar{z}}(\bar{w}) = \frac{d\bar{w}}{d\bar{z}} A_{\bar{z}}(\bar{z})$$

$$\therefore A^{\underbrace{z \dots z}_n}(\bar{z}) = \left(\frac{dw}{dz}\right)^n A^{\underbrace{z \dots z}_n}(w(z))$$

$$A_{\underbrace{\bar{z} \dots \bar{z}}_{\bar{n}}}(\bar{z}) = \left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{n}} A_{\underbrace{\bar{z} \dots \bar{z}}_{\bar{n}}}(\bar{w}(\bar{z}))$$

and likewise $\forall \quad z \rightarrow \bar{z}, \quad w \rightarrow \bar{w}$

$$\therefore \underbrace{A_{\underbrace{\bar{z} \dots \bar{z}}_{\bar{n}}}}_{\substack{\underbrace{}_{\bar{n}} \\ \underbrace{}_n}}(z, \bar{z}) = \left(\frac{dw}{dz}\right)^n \left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{n}} \underbrace{A_{\underbrace{\bar{z} \dots \bar{z}}_{\bar{n}}}}_{\substack{\underbrace{}_{\bar{n}} \\ \underbrace{}_n}}(w(z), \bar{w}(\bar{z}))$$

$$A^{\underbrace{z \dots z}_m}(\bar{z}) = \left(\frac{dw}{dz}\right)^h A^{\bar{z} \dots \bar{z}}(w(z))$$

$h \equiv m - n$ (Conformal weights of A)

$$\star A(z, \bar{z}) = \left(\frac{dw}{dz}\right)^h \left(\frac{d\bar{w}}{d\bar{z}}\right)^{\bar{h}} A(w, \bar{w}) \star$$

Translation: $x' \rightarrow x' + a$ is induced by
 $z \mapsto z + a, \quad \bar{z} \mapsto \bar{z} + a$ for real a

$$z \rightarrow z + a \quad a_n z^{n+1}, \quad n = -1$$

Thus, a general f^h

$$f(z, \bar{z}) \Rightarrow f(z', \bar{z}') = f(z, \bar{z}) + a \partial f + a \bar{\partial} f \\ = f(z, \bar{z}) + a (\partial f + \bar{\partial} f)$$

$$\text{but } L_n = z^{n+1} \partial_z, \quad \bar{L}_n = \bar{z}^{n+1} \bar{\partial}$$

so a General translation:

$$(\text{translation}) = L_{-1} + \bar{L}_{-1}$$

* A shift $x^2 \mapsto x^2 + b$ is induced by
 $z \mapsto z + ib, \quad \bar{z} \mapsto \bar{z} - ib, \quad b \in \mathbb{R}$
(generator) = $L_{-1} - \bar{L}_{-1}$

Rotations: $SO(2) \cong U(1)$

$$\begin{aligned} \delta x^1 &= -\alpha x^2 \\ \delta x^2 &= +\alpha x^1 \end{aligned} \quad \left| \quad \begin{aligned} z &= x^1 + ix^2 \\ \bar{z} &= x^1 - ix^2 \end{aligned} \right.$$

$$z \mapsto (x^1 - \alpha x^2) + i(x^2 + \alpha x^1) = z'$$

$$z' = (x^1 + ix^2) + \alpha(-x^2 + ix^1)$$

$$= (x^1 + ix^2) + \alpha(i^2 x^2 + i x^1)$$

$$= (x^1 + ix^2) + i\alpha(x^1 + x^2)$$

$$= e^{i\alpha} z$$

$$\therefore \boxed{z \mapsto e^{i\alpha} z} \quad \text{and} \quad \boxed{\bar{z} \mapsto e^{-i\alpha} \bar{z}}$$

$$f(z) \mapsto f(z) + \alpha z^{n+1}$$

On a General f^y $f(z, \bar{z})$ a rotation is generated

$$(Rotation) = i\alpha(L_0 - \bar{L}_0)$$

Finally, a dilation $z \mapsto \lambda z, \bar{z} \mapsto \lambda \bar{z}$ is generated by

$$z' = z + (\lambda - 1) z^{0+1}$$

$$\bar{z}' = \bar{z} + (\lambda - 1) \bar{z}^{0+1}$$

$$(Dilation) = L_0 + \bar{L}_0$$

Dilation: $A \mapsto \lambda^{h+\bar{h}} A$
Rotation: $A \mapsto e^{i(h-\bar{h})\alpha} A$

$$\Delta = h + \bar{h} \quad (\text{Dilatation Weight})$$

$$s = h - \bar{h} \quad (\text{Spin})$$

$$\Delta = (m + \bar{m}) - (n + \bar{n})$$

$$s = (m + \bar{n}) - (n + \bar{m})$$

$\leadsto$ Consider fields for which Δ and s are half int.

$$ds^2 = e^\phi dz d\bar{z} \quad \left| \quad \begin{aligned} g_{z\bar{z}} &= \frac{1}{2} e^\phi \\ g^{\bar{z}z} &= 2 e^{-\phi} \end{aligned} \right.$$

$$\rightarrow A_z = g_{z\bar{z}} A^{\bar{z}} = \frac{1}{2} e^\phi A^{\bar{z}}$$

$$A_{\bar{z}} = g_{\bar{z}z} A^z = \frac{1}{2} e^\phi A^z$$

* For Coordinates in $\mathbb{E}^2$: (x^1, x^2)

$$A_z = \frac{\partial x^\alpha}{\partial z} A_\alpha = \frac{\partial x^1}{\partial z} A_1 + \frac{\partial x^2}{\partial z} A_2$$

$$z = x^1 + i x^2 \quad | \quad x^1 = \frac{1}{2}(\bar{z} + z)$$

$$\bar{z} = x^1 - i x^2 \quad | \quad (z - \bar{z}/2i) = x^2$$

$$A_z = \frac{1}{2}(A_1 - iA_2)$$

$$A_{\bar{z}} = \frac{1}{2}(A_1 + iA_2)$$

etc to higher order tensors. $E^2 \rightarrow M^2$

$$A_1 \mapsto A_0 \quad A_{12} \mapsto iA_{01}$$

$$A_2 \mapsto iA_1 \quad A_{21} \mapsto iA_{10}$$

$$A_{11} \mapsto A_{00} \quad A_{22} \mapsto -A_{11}$$

$$* \quad \Gamma_{\beta\gamma}^\alpha = \frac{1}{2} g^{\alpha\sigma} (\partial_\beta g_{\gamma\sigma} + \partial_\gamma g_{\beta\sigma} - \partial_\sigma g_{\beta\gamma})$$

We find $\Gamma = 0$ for all except for

$$\boxed{\Gamma_{zz}^z = \partial_z \varphi} \quad | \quad \boxed{\Gamma_{\bar{z}\bar{z}}^{\bar{z}} = \partial_{\bar{z}} \varphi}$$